

Mehrdad Moradi

✉ mehrdadmoradi@gatech.edu | 🌐 <https://github.com/mehrdadmoradi124> | in themehrdadmoradi | 📍 Atlanta, Georgia, USA

Summary

Mehrdad Moradi is a third-year PhD student in Machine Learning at the Industrial and Systems Engineering Department of Georgia Tech. He is also pursuing a Master's degree in Computer Science with a specialization in machine learning. He has extensive expertise in generative models, with his PhD research centered on their intersection with statistics. His work focuses on applications in computer vision, particularly image-based anomaly detection. His developed methodologies for image-based anomaly detection have been used for additive manufacturing process characterization at the Department of Mechanics at Politecnico di Milano, PV panel smoke and fire detection at the U.S. Department of Energy, and battery capacity prediction at Ford Motor Company.

Education

Georgia Institute of Technology

PhD in Machine Learning (4/4), Advisor: Professor Kamran Paynabar

Atlanta, Georgia

Sep. 2022 - Jun. 2026

Georgia Institute of Technology

MSc in Computer Science (4/4)

Atlanta, Georgia

Sep. 2022 - May. 2025

Sharif University of Technology

B.Sc. in Industrial Engineering (17.6/20), Advisor: Professor Kourosh Eshghi

Tehran, Iran

Sep. 2017 - Feb. 2022

Interests

Diffusion models, Robust Statistics, Anomaly Detection

Publications

- Moradi, M., Paynabar, K. (2025) "RDDPM: Robust Denoising Diffusion Probabilistic Model for Unsupervised Anomaly Segmentation" Submitted to The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025.

Work Experience

Battery Capacity Prediction

Collaboration with Ford Alliance Project, Ford Motor Company

Atlanta, Georgia

Aug. 2024 - Present

Reduced Sim2Real Gap In Robot Learning

Research Intern at Georgia Tech Research Institute (GTRI)

Atlanta, Georgia

May. 2024 - Aug. 2024

- Trained 3dDiffusionPolicy and Robotic View Transformer (RVT) for teleoperated pick and place task in Isaac Lab
- Implemented GenAug data augmentation using depth guided diffusion model for image augmentation
- Implemented MimicGen trajectory augmentation (translation and rotation) on pick and place teleoperated tasks in Isaac Lab
- Reproduced data augmentation module in FoundationPose using open source LLM (Llama3) and Paint3d for 3d texture generation.

Image-based smoke and fire detection in PV panels

Collaboration with United States Department of Energy

Atlanta, Georgia

Aug. 2023 - May. 2024

- Robust principal component analysis (RPCA) for smoke and fire detection

Laser Powder Bed Fusion Process (LPBF) Characterization

Collaboration with Dipartimento di Meccanica, Politecnico di Milano

Atlanta, Georgia

Aug. 2022 - Present

- Developed spatter trajectory detection engine using Kalman filtering
- Energy level prediction accuracy 5% improvement using features extracted from spatter path, size and light (pixel intensity)

Graduate Teaching Assistant

Georgia Institute of Technology

Atlanta, Georgia

Aug. 2022 - May. 2023

- ISyE 3133: Engineering Optimization
 - Held weekly tutoring sessions for 200 undergraduate students in the course.
 - Mentored 3 students on MIP, and Simplex algorithm.

Undergraduate Research Assistant in Dr. Eshghi's Lab

Sharif University of Technology

Tehran, Iran

Jun. 2021 - Feb. 2022

- Capstone Project: Robust Inverse Optimization in Health Care (Passed with honors, achieving a score of 19.5/20)
 - Reviewed literature on tractable reformulation of LP and MIP with different geometric uncertainty sets for their parameters. Uncertainty sets were based on L_1 , L_2 and L_∞ norms.
 - Reviewed literature on tractable reformulation of LP and MLP problems for which we want to find objective function to make a desired solution optimal with minimum distance from current objective function parameters. The distance were based on L_1 and L_∞ norms.
 - Reviewed Robust Inverse Optimization tractable formulations: inverse optimization with uncertainty in desired solution. Uncertainty sets were based on L_2 norm.
 - Applied Robust Inverse Optimization for diet recommendation problem using CPLEX programming with simulated data and illustrated the results.

Professional Presentations

Spatter Trajectory Analysis for Characterizing Laser Powder Bed Fusion Process

Phoenix, Arizona

2023 INFORMS Annual Meeting

Oct. 2023

- Presented my work in the "Advancements in spatial-temporal analytics" session at INFORMS.

Significant Projects

Generative modeling of contaminated data

Atlanta, GA

Paper

Fall 2024

- Extended the denoising score matching algorithm to accommodate noise sampled from Laplacian and exponential family distributions.
- Developed a novel training algorithm combining denoising score matching and contrastive divergence to simultaneously estimate the clean data score and the noise energy from contaminated samples.

RobustCLIP: Robust fine-tuning of CLIP on contaminated data

Atlanta, GA

Paper

Fall 2024

- Conducted extensive experiments with CLIP-ViT-B/32 on the iMet Collection 2019 dataset for the fine-grained recognition of artwork under varying levels of contamination in text descriptions
- Developed a novel L1 norm-based robust loss for RobustCLIP.
- Achieved over 100% improvement in F2 score using RobustCLIP compared to standard CLIP for contaminations under 50%.

Leveraging Large Language Models for Author Attribution in Classical Philosophy Texts

Atlanta, GA

Paper, Code, Data

Spring 2024

- Ran extensive experiments with Llama2 (7b, 13b, 70b), Llama3 (8b), GPT3.5-Turbo.
- Demonstrated preliminary LLM ability in author identification beyond text memorization.
- Showed that few-shot and chain of thought prompting does not improve the results.

Human Drawing Reconstruction From Different Camera Viewpoints

Atlanta, GA

Paper

Spring 2024

- Implemented hand-tracking with hand keypoint detection using Multiview Bootstrapping method.
- Used CNN model for classification of sketches with different views.

Medical Image Analysis Project - Brain Tumor Detection with Deep Learning

Atlanta, GA

Paper, Code, Data

Spring 2023

- Created CNN and RNN models for tumor classification with interpretability using gradient saliency maps and attention-based visualizations.
- Enhanced accuracy by implementing statistical classification algorithms like random forest on pre-trained neural networks feature maps including VAE.
- Achieved an outstanding 80 percents accuracy rate on PCA based classification, and 75 percents on RNN and CNN classifiers.

Optimizing Deep Learning for Medical Image Analysis - Blood Composition Detection

Atlanta, GA

Paper

Spring 2023

- Used pre-trained variants of YOLO object detection model for robust blood cell detection.
- Enhanced memory efficiency (50%) and inference speed (20%) by randomly pruning small weights in the model and 16-bit data precision.
- Explored IOU for duplicate removal and optimizer impact on model performance.
- Visualized convolutional kernels for model interpretability.

Technical Skills

Machine Learning/Data Science Python (Pytorch, TensorFlow, Pandas, Scikit-learn, Matplotlib), R, Matlab, SQL, SAS, Minitab, Access
Optimization GAMS, CPLEX, LINGO

Awards and Honors

ISyE travel fund: 1000\$ reward for conference travel.

Atlanta, Georgia

Georgia Institute of Technology

2023

Topper Fellowship Award: 2500\$ reward for exceptional admitted PhD students.	<i>Atlanta, Georgia</i>
Georgia Institute of Technology	2022
Honorary Tile: Top 5 in GPA among nearly 80 IE undergraduate students.	<i>Tehran, Iran</i>
Sharif University of Technology	2018-2021
INEF scholarship for outstanding academic success.	<i>Tehran, Iran</i>
Iran's National Elites Foundation (INEF)	2017
69th place among 148429 students for mathematics students in national university entrance exam (Konkour).	<i>Tehran, Iran</i>
National Organization for Development of Exceptional Talents (NODET)	2017
Acceptance in the first round of national mathematics and computer Olympiad.	<i>Tehran, Iran</i>
National Organization for Development of Exceptional Talents (NODET)	2014, 2015

Selected Graduate Courses ---

Convex Optimization, Introduction to Uncertainty Quantification, Generative and Geometric Deep Learning, Advanced Topics in Machine Learning, Large Language Models, Advanced Computer Vision, Deep Learning, Linear Optimization, Non-linear Optimization, Mathematical Foundation of Machine Learning, Analysis 1, Computational Science and Engineering Algorithms, Regression Analysis, High Dimensional Data Analytics, Simulation, Transportation Planning, Statistical Machine Learning, Theoretical Statistics